

Query-Efficient Hard-Label Attack: A Prior-Guided Adam Ray Search Optimization

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Abstract

Deep neural networks are vulnerable to adversarial examples, even in hard-label black-box settings where only the top-1 prediction is available. To address the challenges of high-dimensional optimization under limited query budgets, we propose two query-efficient attack methods: AdamOPT, which integrates Adam-based adaptive optimization into the ray-search framework to stabilize and accelerate zeroth-order gradient updates, and PriorAdamOPT, which further incorporates transfer-based priors from surrogate models to enhance gradient estimation. AdamOPT leverages historical gradient information and per-parameter adaptive updates to improve convergence, while PriorAdamOPT constructs a prior-guided orthogonal search basis that combines surrogate and random directions, enhancing both gradient accuracy and query efficiency. Our approach demonstrates superior performance across CIFAR-10, ImageNet, and zero-shot CLIP models, consistently reducing perturbation magnitudes and improving attack efficiency compared to state-of-the-art hard-label attacks. Ablation studies highlight the importance of the number of vectors used for gradient estimation and the quality of surrogate models, showing that combining adaptive optimization with transfer-based priors provides a scalable and robust framework for generating high-quality adversarial examples in challenging black-box scenarios.

Keywords: black-box adversarial attack; Adam optimization; adversarial examples; hard-label attacks; AI security

1. Introduction

Over the past decade, deep neural networks (DNNs) have made remarkable progress in areas such as object recognition, autonomous systems, and face recognition, achieving unprecedented classification accuracy. In particular, deep learning has demonstrated outstanding performance in visual tasks, including image classification, object detection, and image segmentation. However, as these technologies continue to evolve, DNNs face increasingly sophisticated security threats. Research has shown that DNNs are highly vulnerable to carefully crafted adversarial examples, which are generated by adding subtle and often imperceptible perturbations to original images. Despite their minimal magnitude, these perturbations can cause DNNs to produce incorrect predictions or classification results. Early studies on adversarial attacks [1] mainly focused on the white-box setting, where adversaries have full access to the target model's architecture, parameters, and gradient information. Nevertheless, such assumptions are often impractical in real-world applications, motivating growing attention toward black-box attack scenarios, where internal model information, such as parameters and gradients, is inaccessible to the adversary. Among

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them, hard-label attacks are considered one of the most restrictive and challenging settings, since the adversary can access only the final top-1 predicted label without any confidence scores or probability distributions, which is different from typical score-based attacks [2]. This highly limited feedback setting constitutes the primary focus of this paper.

In hard-label attacks, predictions change only near the decision boundary, resulting in a highly discontinuous objective function. Consequently, the attack process involves solving a highly challenging, high-dimensional combinatorial optimization problem. Existing hard-label attack methods typically rely on random walk strategies or gradient estimation to search for adversarial directions. These approaches generally start from a sample with large adversarial perturbations and iteratively reduce distortion by moving along the decision boundary toward the original image. However, such methods often suffer from high query complexity and lack theoretical convergence guarantees. Excessive queries may also increase the risk of being detected by defense or monitoring systems, thereby leading to attack failure. This challenge becomes particularly severe in high-dimensional spaces, such as ImageNet, where the search space for adversarial perturbations is extremely large.

A representative class of hard-label attacks, namely the OPT-series methods [3–6], formulates hard-label attacks as continuous optimization problems. Representative approaches, including OPT [3], Sign-OPT [4], Prior-OPT [5], and PARS-OPT [6] optimize a search direction θ to minimize the ℓ_p -norm distance from a benign image to the adversarial region. For example, OPT employs a randomized gradient-free (RGF) [7] estimator. However, each function evaluation required by finite-difference gradient estimation involves a costly binary search to locate the decision boundary, resulting in substantial query overhead. Sign-OPT [4] reduces query complexity by leveraging the sign of directional derivatives, but at the cost of degraded gradient accuracy. Prior-OPT [5] further improves efficiency by incorporating transfer-based priors into gradient estimation, thereby alleviating the poor estimation quality caused by purely random directions. However, it still relies on simple gradient descent updates and makes limited use of historical gradient information, leading to suboptimal convergence under tight query budgets. Overall, existing methods still largely rely on simple optimization strategies and fail to fully exploit advanced optimization algorithms, such as Adam [8], to accelerate convergence and improve query efficiency.

To address the aforementioned challenges, this paper proposes AdamOPT and its prior-guided variant, PriorAdamOPT. The proposed method employs an Adam-based optimization strategy to improve the convergence and stability of ray-search optimization. By leveraging momentum and adaptive update mechanisms, Adam helps smooth noisy gradient estimates and guide the search direction more effectively. After each update, the direction is normalized to satisfy the unit-norm constraint, ensuring that the optimization is performed over the directional space rather than the perturbation magnitude. Building on this framework, PriorAdamOPT incorporates surrogate-model gradients as transfer-based priors. These priors are combined with random directions and orthogonalized to construct a more informative search basis, thereby improving the accuracy of gradient estimation and enhancing query efficiency under limited query budgets.

The main contributions of this paper are as follows:

- We introduce an Adam-based ray-search optimization strategy that decouples perturbation direction and magnitude. The search direction is updated using Adam with historical gradient information, normalized to satisfy the unit-norm constraint, and then evaluated by binary search to obtain the corresponding boundary distance. This adaptive update mechanism stabilizes noisy zeroth-order gradient estimation and improves convergence and query efficiency under limited query budgets.

- We further improve Adam-based direction optimization by incorporating surrogate-gradient priors into the gradient estimation process. The prior-guided orthogonal basis provides more reliable update directions for Adam, enabling its momentum and adaptive update mechanisms to accumulate useful historical information more effectively. This design enables faster and more stable direction updates, thereby improving query efficiency under limited query budgets.
- Experimental results on standard benchmarks, including CIFAR-10 [9] and ImageNet [10], show that PriorAdamOPT achieves better query efficiency and lower perturbation distortion than existing hard-label attacks, such as Sign-OPT and Prior-OPT.

2. Related Work

Research on geometric optimization and evolutionary strategies has yielded significant advancements. Dong et al. [11] proposed the Evolutionary algorithm, which employs a black-box optimization method based on Covariance Matrix Adaptation Evolution Strategy (CMA-ES) to find optimal solutions by adding random noise and performing local searches. Rahmati et al. [12] proposed GeoDA, which leverages the property that the average curvature of the deep network decision boundary is small near samples. By estimating normals with few queries, it designs a query-efficient ℓ_p -norm attack. Maho et al. [13] proposed SurFree, a black-box adversarial attack algorithm based on geometric principles. By constructing a hyperplane and searching for closer points within it, it avoids expensive gradient proxy estimation. Reza et al. [14] proposed the curvature-aware geometric decision-based black-box attack algorithm CGBA and its variant CGBA-H. This algorithm searches along a semicircular path on a constrained 2D plane, ensuring that boundary points are found without being affected by boundary curvature. Ma et al. [15] proposed Tangent Attack (TA), a geometry-inspired hard-label attack that constructs a virtual hemisphere around the current adversarial point and searches along tangent directions on this hemisphere to efficiently reduce ℓ_2 distortion.

In the realm of optimization and ray search, notable contributions have also been made. The HSJA [16] initializes with target class samples, pushes them towards the boundary via binary search, and estimates the gradient direction, gradually reducing the step size to optimize ℓ_2 and ℓ_∞ distances, significantly improving query efficiency. The ray search attack RayS [17] proposed by Chen and Gu relies solely on hard-label outputs. It performs a binary search on discrete ray directions to find the local optimum of the decision boundary radius, greatly reducing the number of queries. Its hierarchical search version exploits spatial correlations to search on small patches, thereby improving efficiency, and has become one of the baselines for current hard-label attacks.

Additionally, reinforcement learning and adaptive strategies have been integrated into attack methodologies. Yan et al. [18] proposed PDA (Policy Driven Attack), a decision-based black-box attack using reinforcement learning. It samples direction vectors from a normal distribution and performs jump searches based on rewards to select the optimal direction. Li et al. [19] proposed a hard-label black-box attack method, the Adaptive History-driven Attack, which dynamically adjusts based on historical query information. By adaptively optimizing direction coefficients to balance the weights between source and spherical directions, it effectively reduces perturbation magnitudes. Shi et al. [20] propose Patch-wise Adversarial Removal (PAR), a decision-based black-box attack for Vision Transformers that reduces query complexity by iteratively compressing perturbations on less sensitive patches.

3. The Proposed Approach

3.1. Problem Definition of Hard Label Attacks

In the hard-label black-box setting, the adversary has access only to the top-1 predicted label of the target classifier, and the objective is to find a minimal perturbation that induces misclassification. Given a classifier $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^C$ for a C -class classification task, and a benign input sample $\mathbf{x} \in [0, 1]^d$ with ground-truth label y , the attack can be formulated as the following constrained optimization problem:

$$\min_{\mathbf{x}_{\text{adv}}} \|\mathbf{x}_{\text{adv}} - \mathbf{x}\|_p \quad \text{s.t.} \quad \Phi(\mathbf{x}_{\text{adv}}) = 1, \quad (1)$$

where $\Phi(\cdot)$ is the adversarial success indicator function defined as

$$\Phi(\mathbf{x}_{\text{adv}}) := \begin{cases} 1 & \text{if } \hat{y} = y_{\text{adv}} \text{ in a targeted attack,} \\ & \text{or } \hat{y} \neq y \text{ in an untargeted attack,} \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

where $\hat{y}$ is obtained as the class with the largest output logits of ψ on the adversarial example, i.e., $\hat{y} = \arg \max_{i \in \{1, \dots, C\}} \psi(\mathbf{x}_{\text{adv}})_i$. Here, y denotes the ground-truth label of the benign input $\mathbf{x}$, while y_{adv} represents the desired target class in the targeted attack setting.

From an optimization perspective, the objective defined in Eq. (1) seeks the closest adversarial example to the original input. However, in the hard-label black-box setting, neither gradient information nor confidence scores are available, making direct optimization intractable. To address this challenge, we adopt the ray-search formulation introduced by OPT-series methods [3–6], which optimizes the search direction θ to minimize the boundary distance $f(\theta)$ to the adversarial region. This can be formulated as:

$$\min_{\theta \in \mathbb{R}^d \setminus \{0\}} f(\theta), \quad (3)$$

$$\text{where } f(\theta) := \inf \left\{ \lambda > 0 : \Phi\left(\mathbf{x} + \lambda \frac{\theta}{\|\theta\|}\right) = 1 \right\}.$$

When no adversarial point can be found along direction θ , the function value is set to $f(\theta) = +\infty$. Let θ^* denote the optimal direction obtained by solving Eq. (3); the corresponding adversarial example is then given by $\mathbf{x}^* = \mathbf{x} + f(\theta^*) \frac{\theta^*}{\|\theta^*\|}$, where θ^* is the optimal solution obtained from the minimization problem defined in Eq. (3).

3.2. Adam Optimization for Ray Direction Search

3.2.1. Overview

This section introduces AdamOPT, a hard-label black-box attack method that improves the ray-search paradigm with Adam-based adaptive optimization. Under the ray-search formulation, the perturbation is decomposed into a unit search direction and a boundary distance. AdamOPT optimizes the search direction, while the corresponding boundary distance is determined by binary search along the current ray direction.

Unlike traditional methods that rely on simple gradient descent, AdamOPT estimates a zeroth-order gradient by probing boundary-distance variations along randomly sampled orthogonal directions. The estimated gradient is then used by Adam to update the search direction, allowing the optimizer to exploit historical gradient information and adaptive moment estimates to stabilize noisy updates. After each update, the direction is normalized to satisfy the unit-norm constraint, ensuring that the optimization remains in the directional space. Fig. 1 illustrates the framework of our approach. The procedure of AdamOPT is summarized as follows:

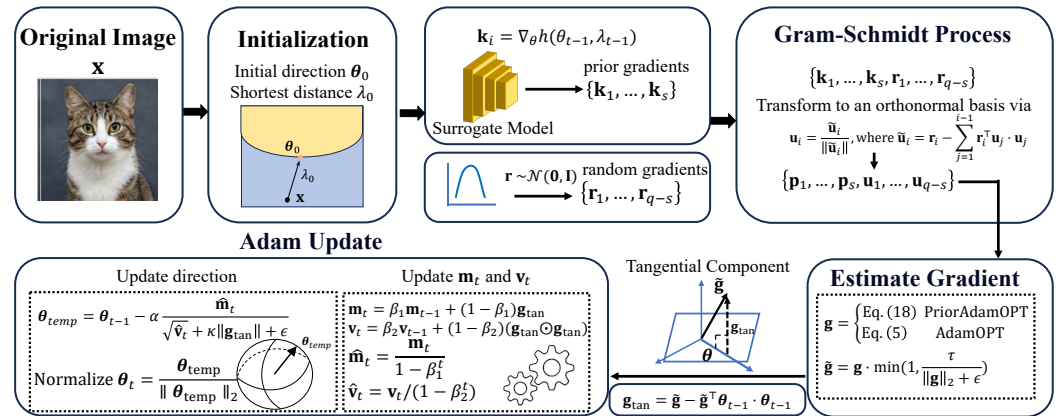


Figure 1. Overview of our framework. After initializing the search direction θ_0 and boundary distance λ_0 , an orthogonal basis is constructed using a Gram-Schmidt process combining surrogate gradients and random directions. Then, the gradient $\mathbf{g}$ is estimated using Eq. (5) and Eq. (18) for AdamOPT and PriorAdamOPT. The estimated gradient is projected onto the tangent space of the current direction θ_{t-1} , and the ray direction θ_t is iteratively updated by using Adam-style update steps. See Algorithm 1 for details.

1. **Initialization:** Search for an initial adversarial direction θ and determine its boundary distance λ via binary search.
2. **Random Basis Construction:** Construct a set of random orthogonal directions $\{\mathbf{u}_i\}_{i=1}^q$ using the Gram-Schmidt process.
3. **Zeroth-order Gradient Estimation:** For each perturbed direction $\theta + \sigma \mathbf{u}_i$, perform binary search to estimate the corresponding boundary distance. The distance variation is used to weight each basis direction and form a zeroth-order gradient estimate.
4. **Adam-based Direction Update:** Update the search direction using Adam with the estimated gradient, apply gradient clipping if needed, and normalize the updated direction to maintain the unit-norm constraint.
5. **Boundary Re-evaluation:** Perform binary search along the updated direction to obtain the new boundary distance, and repeat the process until the query budget or stopping criterion is reached.

3.2.2. Gradient Estimation

In practical hard-label black-box attack scenarios, the gradient and confidence information of the target model are inaccessible. Therefore, AdamOPT estimates the gradient of the objective function $f(\theta)$ using only top-1 predicted labels, where $f(\theta)$ is defined in Eq. (3). Specifically, the algorithm samples q independent random vectors $\{\mathbf{r}_1, \dots, \mathbf{r}_q\}$ from a standard Gaussian distribution and transforms them into an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_q\}$ using the Gram-Schmidt process:

$$\tilde{\mathbf{u}}_i = \mathbf{r}_i - \sum_{j=1}^{i-1} \mathbf{r}_i^\top \mathbf{u}_j \cdot \mathbf{u}_j, \quad \mathbf{u}_i = \frac{\tilde{\mathbf{u}}_i}{\|\tilde{\mathbf{u}}_i\|_2}, \quad i = 1, \dots, q. \quad (4)$$

In AdamOPT, gradient estimation is performed using an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_q\}$, analogous to an orthogonalized variant of Sign-OPT. The key distinction is that AdamOPT constructs the basis via Gram-Schmidt orthonormalization, whereas Sign-OPT relies on randomly sampled Gaussian vectors. Formally, the gradient is estimated as

$$\mathbf{g} := \sum_{i=1}^q \text{sign}(f(\theta + \sigma \mathbf{u}_i) - f(\theta)) \cdot \mathbf{u}_i, \quad (5)$$

where σ is a small non-zero scalar. By leveraging the fact that the sign of a directional derivative can be estimated with a single query [4], each term $\text{sign}(f(\theta + \sigma \mathbf{u}_i) - f(\theta))$ can be computed with one query as follows:

$$\text{sign}(f(\theta + \sigma \mathbf{u}_i) - f(\theta)) = \begin{cases} +1, & \text{if } \Phi\left(\mathbf{x} + f(\theta) \frac{\theta + \sigma \mathbf{u}_i}{\|\theta + \sigma \mathbf{u}_i\|}\right) = 0, \\ -1, & \text{otherwise,} \end{cases} \quad (6)$$

where $\Phi(\cdot)$ is defined in Eq. (2).

To improve optimization stability and suppress excessively large updates caused by noisy finite-difference estimation, we apply gradient clipping to the estimated gradient:

$$\tilde{\mathbf{g}} = \mathbf{g} \cdot \min\left(1, \frac{\tau}{\|\mathbf{g}\|_p + \epsilon}\right), \quad (7)$$

where τ denotes the maximum allowed gradient norm, p is the norm type used for clipping, and ϵ is a small constant for numerical stability.

3.2.3. Adam Optimization of Ray Directions

After obtaining the clipped zeroth-order gradient estimate $\tilde{\mathbf{g}}$, AdamOPT updates the ray direction θ using an Adam-style optimizer. Since the objective $f(\theta)$ depends only on the unit ℓ_2 -norm search direction rather than the magnitude of θ , the optimization is performed on the unit sphere. Therefore, the radial component of the gradient, which is parallel to θ , does not contribute to changing the ray direction. To obtain a valid direction update, we first project the estimated gradient $\tilde{\mathbf{g}}$ onto the tangent space at θ : $\mathbf{g}_{\text{tan}} = \tilde{\mathbf{g}} - \tilde{\mathbf{g}}^\top \theta \cdot \theta$, where $\mathbf{g}_{\text{tan}}$ denotes the tangential gradient on the unit sphere. This projection removes the radial component and ensures that the update only changes the orientation of the ray.

To further enhance optimization efficiency and stability, the algorithm incorporates the adaptive moment estimation of the Adam optimizer. The simplified two-dimensional illustration of our algorithm is shown in Fig. 2.

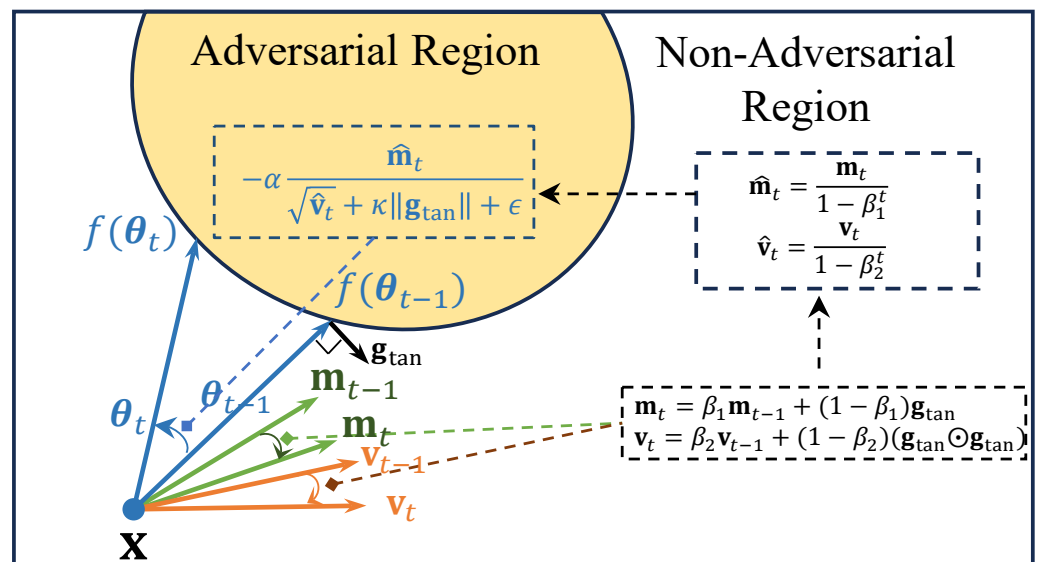


Figure 2. Illustration of the AdamOPT update process. The search direction θ_i is iteratively updated using an Adam-style optimizer, which leverages the tangential component of the zeroth-order gradient $\mathbf{g}_{\text{tan}}$ to compute first- and second-order moments ($\mathbf{m}_i$ and $\mathbf{v}_i$) for adaptive per-parameter updates. The updated direction is then normalized to satisfy the unit-norm constraint on the hypersphere. The dashed boxes show the Adam updates of the first- and second-order moments, which are used to compute the updated ray direction.

The traditional Adam optimizer [21] internally maintains the first-order moment $\mathbf{m}$ (exponential moving average of gradients) and the second-order moment $\mathbf{v}$ (exponential moving average of squared gradients). Inspired by this, our AdamOPT employs the tangential component $\mathbf{g}_{\text{tan}}$ of the estimated gradient $\hat{\mathbf{g}}$ to estimate the first-order and second-order moment. We first update the first-order moment estimate $\mathbf{m}_t$ using an exponential weighted average of the historical gradients and the current gradient, as shown in Eq. (8).

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_{\text{tan}}, \quad (8)$$

where β_1 and β_2 are the exponential decay rates for the first-order and second-order moment estimates, respectively. The first-order moment estimate in Eq. (8) smooths the optimization trajectory by aggregating historical gradients with the current gradient through exponential moving averaging. As a result, AdamOPT is less sensitive to occasional noisy or unstable gradient estimates, while consistently aligned gradients across successive iterations are progressively reinforced, leading to a more stable and reliable optimization direction. Then, we apply Eq. (9) to update the second-order moment $\mathbf{v}_t$.

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) (\mathbf{g}_{\text{tan}} \odot \mathbf{g}_{\text{tan}}). \quad (9)$$

The second-order moment estimate in Eq. (9) adaptively tracks the magnitude of the tangential gradients via exponential averaging of their squared values, allowing AdamOPT to dynamically adjust the effective update scale for each dimension. Specifically, dimensions with consistently large gradients are assigned smaller effective update steps, while dimensions with relatively small gradients receive comparatively larger adaptive updates. This adaptive scaling mechanism improves optimization stability under noisy zeroth-order gradient estimation. The third step is the bias correction, which uses Eqs. (10), (11) to correct the issue of overly small estimates in previous steps.

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}, \quad (10)$$

$$\hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t}. \quad (11)$$

Bias correction compensates for the initialization bias of the moment estimates in early iterations, thereby providing more accurate adaptive update directions and improving optimization stability. Finally, we use Eq. (12) to update the ray direction of AdamOPT:

$$\boldsymbol{\theta}_{\text{temp}} = \boldsymbol{\theta}_{t-1} - \alpha \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t + \kappa \|\mathbf{g}_{\text{tan}}\| + \epsilon}}, \quad (12)$$

where α is the step size, ϵ is a numerical stability constant, and $\kappa \|\mathbf{g}_{\text{tan}}\|$ regulates the update step size and prevents overly large updates on the unit sphere, which may lead to overshooting the local optimum during direction optimization. Since the updated vector $\boldsymbol{\theta}_{\text{temp}}$ generally deviates from the unit sphere, we further enforce the constraint $\|\boldsymbol{\theta}\|_2 = 1$ through normalization:

$$\boldsymbol{\theta}_t = \frac{\boldsymbol{\theta}_{\text{temp}}}{\|\boldsymbol{\theta}_{\text{temp}}\|_2}. \quad (13)$$

In all experiments, we set the hyperparameters as $\alpha = 0.4$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and $\kappa = 0.1$. The above procedure constitutes the core optimization loop for ray-direction search on the unit sphere. Specifically, the tangential update removes the radial component of the gradient to ensure that the optimization focuses only on direction changes, while the normalization step is applied to project the updated $\boldsymbol{\theta}_{\text{temp}}$ back onto the unit sphere.

Built upon this geometry-aware optimization framework, AdamOPT further leverages the adaptive moment estimation mechanism of Adam to achieve more stable and reliable zeroth-order optimization.

3.2.4. Boundary-Distance Estimation via Two-Stage Search

For a given normalized ray direction θ , AdamOPT evaluates the objective function value $f(\theta)$ by estimating the minimum perturbation magnitude required to reach the adversarial region along this direction. Specifically, $f(\theta)$ is defined as

$$f(\theta) = \inf \left\{ \lambda > 0 : \Phi \left(\mathbf{x} + \lambda \frac{\theta}{\|\theta\|} \right) = 1 \right\} \quad (14)$$

where $\Phi(\cdot)$ the attack success indicator function defined in Eq. (2).

This boundary-distance estimation is required whenever AdamOPT evaluates a ray direction. During initialization, after a valid adversarial direction is found, the algorithm refines its boundary distance to obtain the initial objective value $f(\theta_0)$. During iterative optimization, after the direction update to θ_{t+1} , the same procedure is used to estimate the new objective value $f(\theta_{t+1})$. Since confidence scores and gradients of the target model are unavailable, AdamOPT can only determine whether a queried point $\mathbf{x} + \lambda\theta$ lies inside or outside the adversarial region according to its top-1 predicted label.

The search procedure consists of two stages. The first stage constructs an interval $[\lambda_{lo}, \lambda_{hi}]$ that contains the decision boundary. Given an initial distance estimate λ , AdamOPT first checks the prediction of $\mathbf{x} + \lambda\theta$. For untargeted attacks, if this point is still classified as the original label, then λ is regarded as a lower bound, and the algorithm progressively enlarges the upper bound until misclassification occurs. Otherwise, if $\mathbf{x} + \lambda\theta$ is already adversarial, then λ is regarded as an upper bound, and the algorithm progressively decreases the lower bound until the prediction returns to the original label. For targeted attacks, the same bracketing strategy is applied using the target-label condition: the upper bound is enlarged until the target class is reached, or the lower bound is reduced while the perturbed image remains classified as the target class.

After a valid interval is obtained, AdamOPT performs binary search within $[\lambda_{lo}, \lambda_{hi}]$ to refine the boundary-distance estimate. At each iteration, the midpoint

$$\lambda_{mid} = \frac{\lambda_{lo} + \lambda_{hi}}{2} \quad (15)$$

is evaluated by querying the prediction of $\mathbf{x} + \lambda_{mid}\theta$. For untargeted attacks, if the midpoint already causes misclassification, it becomes the new upper bound; otherwise, it becomes the new lower bound. For targeted attacks, if the midpoint is classified as the target label, it becomes the new upper bound; otherwise, it becomes the new lower bound. This process is repeated until the interval length is smaller than a prescribed tolerance, and the final upper bound is used as the estimated value of $f(\theta)$.

In practice, different tolerances can be used for different purposes. A relatively coarse tolerance is sufficient when the boundary search is used inside zeroth-order gradient estimation, while a finer tolerance can be adopted when evaluating the updated ray direction. To avoid degenerate search behavior, AdamOPT further includes numerical safeguards, such as stopping the search when the interval can no longer be refined due to floating-point precision or when the number of search iterations exceeds a predefined limit. These mechanisms provide a stable and query-efficient estimate of the ray-based objective function value under hard-label feedback.

3.3. Prior-guided Adam Optimization for Ray Direction Search

Although AdamOPT benefits from momentum and adaptive optimization, the gradient estimation still relies on purely random exploration directions, without exploiting more informative transfer-based priors from surrogate models. Consequently, their performance is suboptimal even under a budget of 2,000 queries. To address this, we introduce a transfer-based prior into AdamOPT, resulting PriorAdamOPT algorithm.

Unlike AdamOPT, PriorAdamOPT leverages surrogate models to provide more informative transfer-based priors, thereby enhancing gradient estimation. We assume that the adversary has access to several white-box surrogate models, from which gradients can be computed via backpropagation. However, a key challenge remains: even for white-box models, the gradient of the objective function $f(\theta)$, defined in Eq. (14), cannot be directly obtained, as it is evaluated via the non-differentiable binary search procedure described in Section 3.2.4. To address this issue, we adopt a differentiable surrogate function $h(\theta, \lambda)$, following Ma et al. [5], defined as

$$h(\theta, \lambda) := \begin{cases} \hat{\psi}_y - \max_{j \neq y} \hat{\psi}_j, & \text{for untargeted attacks,} \\ \max_{j \neq y_{\text{adv}}} \hat{\psi}_j - \hat{\psi}_{y_{\text{adv}}}, & \text{for targeted attacks,} \end{cases} \quad (16)$$

where $\hat{\psi}_i := \hat{\psi}(\mathbf{x} + \lambda \cdot \frac{\theta}{\|\theta\|})_i$ denotes the i -th element of the surrogate model's output $\hat{\psi}$, $\mathbf{x}$ is the original image, and y and y_{adv} represent the true label and target class label, respectively. The surrogate function $h(\theta, \lambda)$ effectively corresponds to the negative Carlini & Wagner (C&W) loss evaluated at the surrogate model. Its key property is that, for any non-zero vector $\theta_0 \in \mathbb{R}^d$ with finite objective $f(\theta_0) < +\infty$, the gradient of the true objective function $f(\theta)$ is proportional to the gradient of the surrogate:

$$\nabla f(\theta_0) = c \cdot \nabla_{\theta} h(\theta_0, \lambda_0), \quad (17)$$

where $\lambda_0 = f(\theta_0)$ is treated as a constant during differentiation, and c is a non-zero scalar. The specific value λ_0 is obtained via binary search along the ray defined by θ_0 , such that $h(\theta_0, \lambda_0)$ measures the surrogate loss at the decision boundary. The proof of Eq. (17) is provided in Ma et al. [5].

Suppose we have s surrogate models, from which we obtain s prior directions via $\nabla_{\theta} h(\theta_0, \lambda_0)$. We denote these prior directions as $\mathbf{k}_1, \dots, \mathbf{k}_s$. In addition, we randomly sample $q - s$ Gaussian vectors $\mathbf{r}_1, \dots, \mathbf{r}_{q-s}$ to complement the prior space.

We then apply the Gram-Schmidt process with Eq. (4) to orthogonalize and normalize the combined set of vectors $\mathbf{k}_1, \dots, \mathbf{k}_s, \mathbf{r}_1, \dots, \mathbf{r}_{q-s}$, resulting in an orthonormal basis $\mathbf{p}_1, \dots, \mathbf{p}_s, \mathbf{u}_1, \dots, \mathbf{u}_{q-s}$. Here, $\mathbf{p}_1, \dots, \mathbf{p}_s$ correspond to the prior directions $\mathbf{k}_1, \dots, \mathbf{k}_s$, while $\mathbf{u}_1, \dots, \mathbf{u}_{q-s}$ correspond to the random directions $\mathbf{r}_1, \dots, \mathbf{r}_{q-s}$. Our objective is then to estimate the projection of the gradient onto the subspace S spanned by this orthonormal basis using as few queries as possible. In PriorAdamOPT, we adopt a gradient estimation procedure of Prior-OPT [5]. Specifically, the gradient is approximated as

$$\mathbf{g} := \sum_{i=1}^s \frac{f(\theta + \sigma \mathbf{p}_i) - f(\theta)}{\sigma} \cdot \mathbf{p}_i + \frac{f(\theta + \sigma \overline{\mathbf{g}}_{\perp}) - f(\theta)}{\sigma} \cdot \overline{\mathbf{g}}_{\perp}, \quad (18)$$

where $\overline{\mathbf{g}}_{\perp}$ denotes the ℓ_2 -normalized vector of $\mathbf{g}_{\perp}$, and $\mathbf{g}_{\perp}$ is computed as

$$\mathbf{g}_{\perp} := \sum_{i=1}^{q-s} \text{sign}(f(\theta + \sigma \mathbf{u}_i) - f(\theta)) \cdot \mathbf{u}_i, \quad (19)$$

where each term $\text{sign}(f(\theta + \sigma \mathbf{u}_i) - f(\theta))$ can be estimated with one query by using Eq. (6).

Once the estimated gradient $\mathbf{g}$ is obtained, we first apply the gradient clipping technique in Eq. (7) to obtain the clipped gradient $\tilde{\mathbf{g}}$. We then project $\tilde{\mathbf{g}}$ onto the tangent space at $\boldsymbol{\theta}$: $\mathbf{g}_{\text{tan}} = \tilde{\mathbf{g}} - (\tilde{\mathbf{g}}^\top \boldsymbol{\theta})\boldsymbol{\theta}$, where $\mathbf{g}_{\text{tan}}$ denotes the tangential gradient on the unit sphere.

Subsequent updates follow the same procedure as in AdamOPT, i.e., we perform Adam-based updates according to Eqs. (8), (9), (10), (11), (12), and (13). The complete workflow of AdamOPT and PriorAdamOPT is summarized in Algorithm 1.

Algorithm 1 AdamOPT and PriorAdamOPT

- 1: **Input:** the original image $\mathbf{x}$, the success indicator function $\Phi(\cdot)$, number of estimation vectors q , numerical stability constant ϵ , step size α , first-order moment decay β_1 , second-order moment decay β_2 , tangent step scaling κ , number of iterations T , maximum gradient norm τ , surrogate model set $\mathbb{S} = \{\hat{\psi}_1, \dots, \hat{\psi}_s\}$ with $s > 0$ for PriorAdamOPT, and $\mathbb{S} = \emptyset$ for AdamOPT.
 - 2: **Output:** adversarial example $\mathbf{x}^*$ that satisfies $\Phi(\mathbf{x}^*) = 1$.
 - 3: Generate 100 candidate images Ω by adding random-direction perturbations to $\mathbf{x}$;
 - 4: $\tilde{\mathbf{x}}_0 \leftarrow \begin{cases} \arg \min_{\tilde{\mathbf{x}}_0 \in \Omega} \|\tilde{\mathbf{x}}_0 - \mathbf{x}\|_2 & \text{s.t. } \Phi(\tilde{\mathbf{x}}_0) = 1, \\ \text{randomly select a target-class image } \tilde{\mathbf{x}}_0 \text{ satisfying } \Phi(\tilde{\mathbf{x}}_0) = 1, \end{cases}$ untargeted attack
targeted attack
 - 5: $\boldsymbol{\theta}_0 \leftarrow \frac{\tilde{\mathbf{x}}_0 - \mathbf{x}}{\|\tilde{\mathbf{x}}_0 - \mathbf{x}\|_2}$, $d_0 \leftarrow \|\tilde{\mathbf{x}}_0 - \mathbf{x}\|_2$;
 - 6: $\mathbf{m}_0 \leftarrow \mathbf{0}$, $\mathbf{v}_0 \leftarrow \mathbf{0}$; ► Initialize first-order and second-order moments.
 - 7: **for** t in $1, \dots, T$ **do**
 - 8: **for** $\hat{\psi}_i$ in $\mathbb{S}$ **do**
 - 9: $\lambda_{t-1} \leftarrow \text{BinarySearch}(\mathbf{x}, \boldsymbol{\theta}_{t-1}, \hat{\psi}_i, \Phi)$;
 - 10: $\mathbf{k}_i \leftarrow \nabla_{\boldsymbol{\theta}} h(\boldsymbol{\theta}_{t-1}, \lambda_{t-1})$ on $\hat{\psi}_i$ with λ_{t-1} treated as a constant in differentiation; ►
 Obtain s transfer-based priors and $h(\cdot, \cdot)$ is defined in Eq. (16).
 - 11: **end for**
 - 12: $\mathbf{r}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $i = 1, \dots, q - s$;
 - 13: $\mathbf{p}_1, \dots, \mathbf{p}_s, \mathbf{u}_1, \dots, \mathbf{u}_{q-s} \leftarrow \text{Gram-Schmidt}(\mathbf{k}_1, \dots, \mathbf{k}_s, \mathbf{r}_1, \dots, \mathbf{r}_{q-s})$;
 - 14: Estimate gradient $\mathbf{g} \leftarrow \begin{cases} \text{Eq. (18)}, & \text{if PriorAdamOPT} \\ \text{Eq. (5)}, & \text{if AdamOPT} \end{cases}$
 - 15: $\tilde{\mathbf{g}} \leftarrow \mathbf{g} \cdot \min\left(1, \frac{\tau}{\|\mathbf{g}\|_2 + \epsilon}\right)$; ► Clip the gradient with the maximum norm τ .
 - 16: $\mathbf{g}_{\text{tan}} \leftarrow \tilde{\mathbf{g}} - \tilde{\mathbf{g}}^\top \boldsymbol{\theta}_{t-1} \cdot \boldsymbol{\theta}_{t-1}$ ► Compute the tangential gradient.
 - 17: $\mathbf{m}_t \leftarrow \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_{\text{tan}}$; ► First-order moment.
 - 18: $\mathbf{v}_t \leftarrow \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) (\mathbf{g}_{\text{tan}} \odot \mathbf{g}_{\text{tan}})$; ► Second-order moment.
 - 19: $\hat{\mathbf{m}}_t \leftarrow \frac{\mathbf{m}_t}{1 - \beta_1^t}$, $\hat{\mathbf{v}}_t \leftarrow \frac{\mathbf{v}_t}{1 - \beta_2^t}$; ► Bias correction.
 - 20: $\boldsymbol{\theta}_{\text{temp}} \leftarrow \boldsymbol{\theta}_{t-1} - \alpha \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t + \kappa \|\mathbf{g}_{\text{tan}}\| + \epsilon}}$; ► Update direction.
 - 21: $\boldsymbol{\theta}_t \leftarrow \frac{\boldsymbol{\theta}_{\text{temp}}}{\|\boldsymbol{\theta}_{\text{temp}}\|_2}$; ► Normalize direction.
 - 22: $d_t \leftarrow \|f(\boldsymbol{\theta}_t) \cdot \boldsymbol{\theta}_t\|_2$;
 - 23: **end for**
 - 24: **return** $\mathbf{x}^* \leftarrow \mathbf{x} + f(\boldsymbol{\theta}_T) \frac{\boldsymbol{\theta}_T}{\|\boldsymbol{\theta}_T\|_2}$;
-

4. Experiment

4.1. Experimental Setting

4.1.1. Implementation Details and Datasets

In this paper, all experiments are implemented using the PyTorch 1.7.1 framework on NVIDIA V100 GPUs. We conducted experiments on two standard datasets: CIFAR-10 [9] and ImageNet [10]. The image dimensions are $32 \times 32 \times 3$ for CIFAR-10, and $299 \times 299 \times 3$ or $224 \times 224 \times 3$ for ImageNet, depending on the target model architecture. For the experiments, we randomly selected 1,000 images from the validation set. The hyperparameters used in the experiments are shown in the Table 1.

Table 1. Hyperparameter Settings

Hyperparameter	Description	Default Value
α	step size	0.4
q	number of vectors for gradient estimation	200
tol	binary search stopping threshold	1×10^{-3}
Q	maximum number of queries	2,000
δ	attack success threshold	$\sqrt{0.0015 \times d}$
β_1	first-order moment decay	0.9
β_2	second-order moment decay	0.999
κ	tangent step scaling	0.1

4.1.2. Compared Methods

To validate the effectiveness of our proposed approach, we select state-of-the-art baseline methods for comparison, including HopSkipJumpAttack (HSJA) [16] and Tangent Attack (TA) [22], Sign-OPT [4], Evolutionary [11], GeoDA [12], SurFree [13], AHA [19], Biased Boundary Attack (BBA) [23], Prior-Sign-OPT [5] and Prior-OPT [5].

4.1.3. Evaluation Metric

To ensure a fair comparison, all approaches are evaluated using the mean distortion over 1,000 images, defined as: $\frac{1}{|X|} \sum_{\mathbf{x} \in X} \|\mathbf{x}_{\text{adv}} - \mathbf{x}\|_2$, where X denotes the test set. The evaluation is conducted under different query budgets, ranging from 400 to 2,000. Additionally, our experiments report the attack success rate (ASR) under specific query budgets, which is defined as the percentage of samples with distortion lower than the threshold of $\sqrt{0.0015 \times d}$, where d is the image dimension.

4.2. Comparison with State-of-the-Art Attacks

On the CIFAR-10 dataset, we use ResNet-110 as the surrogate model for BBA, Prior-Sign-OPT, Prior-OPT, and our PriorAdamOPT. On the ImageNet dataset, for surrogate-model-based methods (BBA, Prior-Sign-OPT, Prior-OPT, and PriorAdamOPT), we employ ResNet-50 as the surrogate for the ResNet-101 target model, and ConViT [24] as the surrogate for the Swin Transformer [25] target model.

4.2.1. Results of Attacks against Undefended Models

Table 2 reports the mean ℓ_2 distortions of untargeted attacks on the ImageNet dataset across varying query budgets. While BBA achieves the lowest distortion for smaller budgets (under 1K queries), PriorAdamOPT consistently outperforms all baselines when the query budget exceeds 1K, achieving the smallest distortions overall. The standard Adam optimizer converges quickly because it adaptively scales the learning rate for each parameter using estimates of both the first and second moments of the gradients. This per-parameter adaptation allows larger steps in directions of low variance and smaller steps in directions of high variance, stabilizing updates and accelerating convergence.

Our PriorAdamOPT leverages these advantages while further incorporating transfer-based priors from surrogate models, resulting in significantly lower ℓ_2 distortions than Prior-OPT under the same query budgets. This demonstrates that combining a more sophisticated optimization algorithm with transfer-based prior information can substantially improve attack efficiency. Notably, for both ResNet-101 and Swin Transformer target models, PriorAdamOPT achieves consistently smaller distortions for most query levels compared to classical hard-label attacks and surrogate-model-based baselines. These results highlight the effectiveness of our approach in generating high-quality adversarial examples with

Table 2. Mean ℓ_2 distortions across query budgets on ImageNet.

	Method	ℓ_2 -Norm Attacks				
		@0.4K	@0.8K	@1.2K	@1.6K	@2.0K
ResNet-101	HSJA [16]	46.073	32.457	27.552	20.909	18.389
	TA [22]	43.970	31.725	27.355	20.919	18.491
	Sign-OPT [4]	55.486	41.447	32.677	25.269	21.235
	Evolutionary [11]	45.459	35.757	29.421	24.905	21.507
	GeoDA[12]	27.416	22.047	17.968	14.871	12.526
	SurFree [13]	33.594	27.029	22.984	20.183	18.100
	AHA [19]	49.025	38.763	31.987	27.021	23.180
	BBA [23]	26.616	16.431	13.939	12.637	11.764
	Prior-Sign-OPT [5]	55.825	38.644	30.740	23.597	18.733
	Prior-OPT [5]	55.440	22.623	14.812	9.885	7.100
	AdamOPT (Ours)	55.142	43.426	33.660	24.491	19.149
	PriorAdamOPT (Ours)	55.808	19.769	8.163	4.804	4.034
Swin Transformer	HSJA [16]	95.372	74.871	66.059	51.560	45.855
	TA [22]	91.369	73.236	65.470	52.060	46.728
	Sign-OPT [4]	118.726	94.236	77.015	63.140	53.399
	Evolutionary [11]	89.621	74.903	64.182	55.829	49.243
	GeoDA [12]	66.136	57.224	49.334	42.601	36.925
	SurFree[13]	64.720	51.767	43.783	38.224	34.281
	AHA [19]	89.964	72.965	61.326	53.018	46.756
	BBA [23]	66.851	47.626	38.510	33.108	29.375
	Prior-Sign-OPT [5]	117.684	88.613	72.624	58.167	46.623
	Prior-OPT [5]	118.599	52.705	35.009	24.731	19.482
	AdamOPT (Ours)	118.493	94.701	76.718	59.066	47.974
	PriorAdamOPT (Ours)	118.814	48.204	21.113	15.276	13.497

Table 3. Mean ℓ_2 distortions across query budgets on CIFAR-10, where the surrogate model is adopted as ResNet-110.

	Method	ℓ_2 -Norm Attacks				
		@0.4K	@0.8K	@1.2K	@1.6K	@2.0K
WRN-40-10 (with dropout)	HSJA [16]	1.830	0.973	0.779	0.564	0.499
	TA [22]	1.868	0.981	0.779	0.560	0.495
	Sign-OPT [4]	4.667	3.495	2.949	2.337	1.827
	Evolutionary[11]	2.683	1.801	1.343	1.063	0.879
	GeoDA [12]	2.721	1.972	1.615	1.435	1.292
	SurFree [13]	2.141	1.557	1.247	1.061	0.938
	AHA [19]	3.644	2.720	2.176	1.794	1.516
	BBA [23]	2.024	1.034	0.803	0.712	0.655
	Prior-Sign-OPT [5]	4.773	2.921	1.805	1.119	0.803
	Prior-OPT [5]	4.199	2.322	1.473	0.837	0.572
	AdamOPT (Ours)	9.525	6.925	2.779	0.829	0.619
	PriorAdamOPT (Ours)	9.075	6.737	1.805	0.536	0.438

minimal perturbation, emphasizing that optimization strategy and prior knowledge jointly contribute to superior attack performance.

Table 4. The experimental results of attacking against CLIP with the backbone of ViT-L/14.

Method	Mean ℓ_2 Distortions					Attack Success Rate ¹				
	@0.4K	@0.8K	@1.2K	@1.6K	@2.0K	@0.4K	@0.8K	@1.2K	@1.6K	@2.0K
HSJA [16]	67.520	66.726	65.693	64.207	63.363	7.1%	8.1%	8.4%	9.2%	9.8%
Sign-OPT [4]	65.846	60.060	55.929	52.019	49.435	14.9%	16.0%	16.7%	17.6%	17.9%
Evolutionary [4]	49.470	36.408	28.292	22.775	18.810	9.7%	19.2%	30.1%	39.7%	48.8%
TA [22]	61.201	59.446	58.651	57.751	56.892	7.7%	9.1%	9.4%	10.2%	10.6%
SurFree [13]	31.051	23.299	18.809	15.909	13.813	29.9%	44.1%	53.5%	60.8%	66.7%
AHA [19]	54.326	41.682	33.057	27.305	22.925	7.3%	12.2%	20.9%	29.7%	37.5%
BBA [23]	36.840	23.288	17.652	14.592	12.656	9.9%	36.9%	56.6%	66.4%	72.5%
Prior-Sign-OPT [5]	65.461	57.747	51.486	47.111	43.327	14.4%	15.8%	17.3%	19.1%	20.4%
Prior-OPT [5]	65.846	38.044	28.580	23.930	21.267	14.8%	28.7%	39.5%	46.1%	51.6%
AdamOPT (Ours)	66.653	53.657	40.668	28.389	21.610	14.3%	20.1%	26.7%	38.9%	49.8%
PriorAdamOPT (Ours)	65.444	32.963	19.354	14.268	12.260	14.6%	38.7%	58.3%	68.8%	75.5%

¹ The distortion threshold for the attack success rate is 15.026, which is calculated as $\sqrt{0.0015 \times 224 \times 224 \times 3}$.

Table 3 reports the mean ℓ_2 distortions of untargeted attacks on CIFAR-10 with WRN-40 as the target model and ResNet-110 as the surrogate model. PriorAdamOPT achieves the lowest distortions when the query budget exceeds 1K, outperforming all baselines, including Prior-OPT and Prior-Sign-OPT. By combining Adam's adaptive per-parameter updates with transfer-based priors, PriorAdamOPT generates high-quality adversarial examples with smaller perturbations, highlighting the effectiveness of integrating advanced optimization with surrogate model knowledge.

To evaluate the scalability of our proposed approach, we conduct untargeted attacks on the CLIP model with the ViT-L/14 backbone. Table 4 reports the mean ℓ_2 distortions and attack success rates under varying query budgets. PriorAdamOPT consistently achieves the lowest distortions across most query budgets, outperforming both standard AdamOPT and surrogate-model-based baselines such as Prior-OPT and Prior-Sign-OPT. In particular, at higher query budgets such as 2K, PriorAdamOPT reduces the distortion to 12.318 while reaching a success rate of 74.9%, demonstrating its effectiveness in generating high-quality adversarial examples with minimal perturbation. These results highlight that incorporating the surrogate model as a prior substantially improves attack performance, even though the surrogate model is pretrained on ImageNet and differs fundamentally from CLIP's contrastive language-image pretraining. The ability of PriorAdamOPT to leverage knowledge from surrogate models of different architectures underscores the benefit of combining adaptive optimization with transfer-based priors, enabling effective attacks against zero-shot classifiers like CLIP.

4.2.2. Results of Attacks against Defense Models

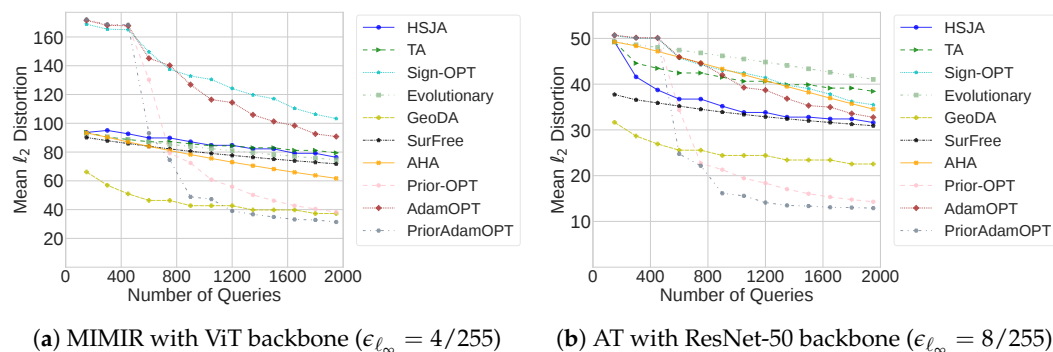


Figure 3. Mean distortions of ℓ_2 -norm untargeted attacks against defense models. The surrogate model of PriorAdamOPT and Prior-OPT is AT with ResNet-50 backbone ($\epsilon_{\ell_\infty} = 4/255$).

We evaluate untargeted attacks against two types of defense models: adversarial training (AT) [26] and MIMIR [27], which achieves state-of-the-art robustness on RobustBench [28]. Fig. 3 presents the mean ℓ_2 distortions of different attack methods on ImageNet under varying query budgets. For early-stage queries, PriorAdamOPT exhibits comparable or slightly higher distortions than some baseline methods, such as GeoDA or Prior-OPT, reflecting its conservative updates at the beginning. However, as the number of queries increases beyond 2,000, PriorAdamOPT's adaptive optimization coupled with transfer-based priors enables it to progressively reduce the distortion, ultimately surpassing all other baselines. This indicates that the full potential of PriorAdamOPT is realized in the later stage of the attack process, where its ability to leverage surrogate knowledge and perform per-parameter adaptive updates allows for more precise and efficient perturbation, achieving superior performance against both AT and MIMIR defense models. Future experiments extending the query budget up to 10,000 will further illustrate the sustained advantage of PriorAdamOPT over existing methods.

4.3. Ablation Studies of PriorAdamOPT

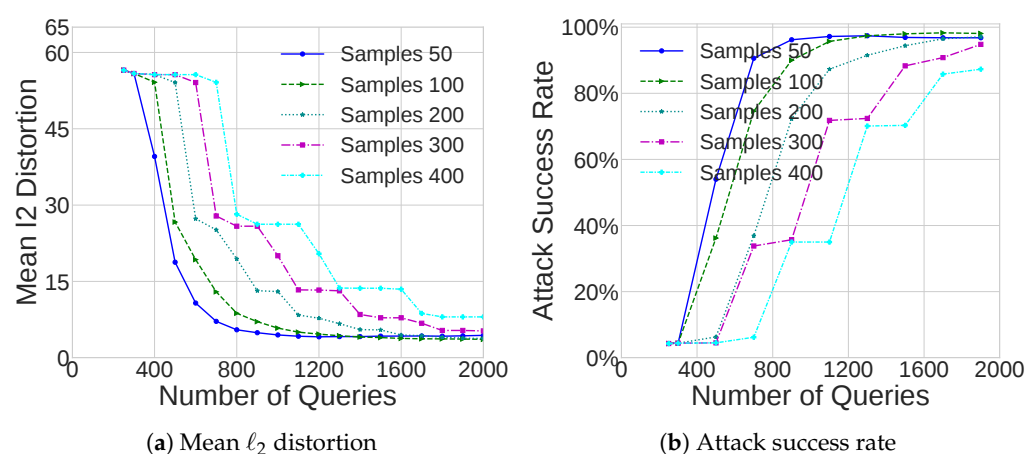


Figure 4. Effect of the number of vectors q used for gradient estimation. Experiments are conducted using PriorAdamOPT for untargeted attacks on ResNet-101, with ResNet-50 as the surrogate model on ImageNet.

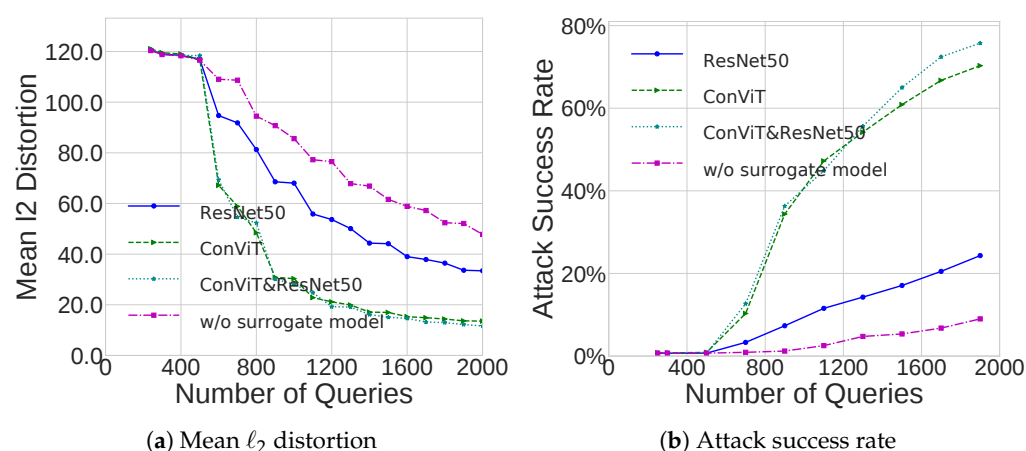


Figure 5. Impact of surrogate model choice. We conduct untargeted attacks on Swin Transformer using PriorAdamOPT, with AdamOPT used for results without a surrogate model, on ImageNet.

Figs. 4 and 5 present ablation studies of PriorAdamOPT, examining two factors: the number of vectors used for gradient estimation and the choice of surrogate models.

The number of vectors used for gradient estimation, denoted as q , balances two competing objectives. If q is too large, gradient estimates become more accurate, but the query budget is quickly depleted; if q is too small, the estimated gradients are unreliable, making the attack more sensitive to discrepancies between the surrogate and target models. To identify an appropriate trade-off, we conducted ablation experiments with varying values of q , as shown in Fig. 4. Based on these results, we set $q = 200$ in our main experiments, which provides a robust compromise between gradient precision and query efficiency.

Similarly, the number and quality of surrogate models directly affect the effectiveness of the prior directions $\mathbf{p}_1, \dots, \mathbf{p}_s$. Surrogate models with higher architectural similarity to the target provide more informative priors, improving search efficiency, whereas dissimilar or low-quality surrogates may mislead the search. Fig. 5 shows that surrogate architectures more similar to the target yield better attack performance, and using multiple surrogate models further enhances performance compared to using a single surrogate model or none at all. Even when a surrogate model (ResNet-50) differs substantially from a Swin Transformer target model, its contribution is weaker than that of ConViT, which is more similar, but it still improves over the baseline without any surrogate model. These results highlight that both the number and quality of surrogate models are crucial for guiding PriorAdamOPT to generate adversarial examples efficiently.

5. Conclusion

In this work, we proposed two query-efficient hard-label black-box attack methods: AdamOPT and its prior-guided variant PriorAdamOPT. AdamOPT introduces an Adam-based adaptive optimization strategy to the ray-search framework, leveraging historical gradient information and per-parameter moment estimates to stabilize zeroth-order gradient updates and accelerate convergence. Building upon this, PriorAdamOPT incorporates transfer-based priors from surrogate models into the gradient estimation process, constructing an orthogonal basis that combines prior and random directions to further improve gradient accuracy and query efficiency. Extensive experiments on CIFAR-10, ImageNet, and zero-shot CLIP classifiers demonstrate that both methods achieve competitive performance, with PriorAdamOPT consistently outperforming AdamOPT and existing baselines in terms of ℓ_2 distortion and attack success rate, particularly under larger query budgets. Ablation studies highlight the importance of selecting an appropriate number of vectors for estimating gradients and high-quality surrogate models. Overall, our results indicate that combining adaptive optimization with transfer-based priors enables more precise and efficient generation of adversarial examples, providing a robust framework for hard-label black-box attacks in high-dimensional and real-world scenarios.

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